Thakur Shree DPS College of Engineering & Management, Vasai (East)

First Year Engineering

Academic Year: 2026-2027

Assignment No: 03

Subject: BSC102/AP **Date: 17/10/2026**
Max Marks: 10

Question 1 [2 Marks]

Find the thickness of the soap film which appears yellow (wavelength 5896 Å) in reflection when it is illuminated by white light at an angle of 45°. Given refractive index of the film is 1.33.

Solution:

- **Given:** $\lambda = 5896$ Å $= 5.896 \times 10^{-5}$ cm, $\mu = 1.33$, angle of incidence $i = 45°$.
- **Angle of refraction (r):**

$$\sin r = \frac{\sin i}{\mu} = \frac{\sin 45°}{1.33} = \frac{0.7071}{1.33} \approx 0.5317$$

$$\cos r = \sqrt{1 - \sin^2 r} = \sqrt{1 - (0.5317)^2} \approx 0.8469$$

- **Condition for bright fringe (maxima in reflection):**

$$2\mu t \cos r = \frac{(2m+1)\lambda}{2}$$

For the minimum non-zero thickness ($m = 0$):

$$2\mu t \cos r = \frac{\lambda}{2} \implies t = \frac{\lambda}{4\mu \cos r}$$

- **Calculation:**

$$t = \frac{5.896 \times 10^{-5}}{4 \times 1.33 \times 0.8469} = \frac{5.896 \times 10^{-5}}{4.5055} \approx 1.309 \times 10^{-5} \text{ cm} = 130.9 \text{ nm}$$

Answer: The thickness of the soap film is $t \approx 1.309 \times 10^{-5}$ cm (or 130.9 nm).

Question 2 [2 Marks]

A soap film 4×10^{-5} cm thick is viewed at an angle of $35°$ to normal. Calculate the wavelength of light in the visible spectrum that will be absent from the reflected light ($\mu = 1.33$).

Solution:

- **Given:** $t = 4 \times 10^{-5}$ cm, angle of incidence $i = 35°$, $\mu = 1.33$.
- **Angle of refraction (r):**

$$\sin r = \frac{\sin 35°}{1.33} = \frac{0.5736}{1.33} \approx 0.4313$$

$$\cos r = \sqrt{1 - \sin^2 r} = \sqrt{1 - (0.4313)^2} \approx 0.9022$$

- **Condition for absent wavelength (minima in reflection):**

$$2\mu t \cos r = m\lambda \implies \lambda = \frac{2\mu t \cos r}{m}$$

$$2\mu t \cos r = 2 \times 1.33 \times (4 \times 10^{-5}) \times 0.9022 \approx 9.599 \times 10^{-5} \text{ cm} = 959.9 \text{ nm}$$

- **Visible spectrum range (400 nm $\leq \lambda \leq$ 700 nm):**

$$\text{For } m = 1: \quad \lambda_1 = 959.9 \text{ nm} \quad \text{(Infrared)}$$

$$\text{For } m = 2: \quad \lambda_2 = \frac{959.9}{2} \approx 480 \text{ nm} \quad \text{(Visible region – Blue)}$$

$$\text{For } m = 3: \quad \lambda_3 = \frac{959.9}{3} \approx 320 \text{ nm} \quad \text{(Ultraviolet)}$$

Answer: The wavelength absent from the reflected light in the visible spectrum is $\lambda \approx 480$ nm (4.8×10^{-5} cm).

Question 3 [2 Marks]

In Newton's Ring experiment, the wavelength of light incident is 5×10^{-5} cm. If the diameter of the 10th dark ring is 0.5 cm, calculate the radius of curvature R.

Solution:

- **Given:** $\lambda = 5 \times 10^{-5}$ cm, ring order $n = 10$, diameter $D_{10} = 0.5$ cm.
- **Formula for dark rings:**

$$D_n^2 = 4nR\lambda \implies R = \frac{D_n^2}{4n\lambda}$$

- **Calculation:**

$$R = \frac{(0.5)^2}{4 \times 10 \times (5 \times 10^{-5})} = \frac{0.25}{200 \times 10^{-5}} = \frac{0.25}{2 \times 10^{-3}} = 125 \text{ cm}$$

Answer: The radius of curvature is $R = 125$ cm (or 1.25 m).

Question 4 [2 Marks]

In Newton's ring experiment, the diameter of the 15th ring was found to be 0.590 cm and that of the 5th ring was 0.336 cm. If the radius of the plano-convex lens is 100 cm, compute the wavelength of light used.

Solution:

- **Given:** $D_{n+p} = D_{15} = 0.590$ cm, $D_n = D_5 = 0.336$ cm, $p = 15 - 5 = 10$, $R = 100$ cm.
- **Formula:**

$$\lambda = \frac{D_{n+p}^2 - D_n^2}{4pR}$$

- **Calculation:**

$$D_{15}^2 = (0.590)^2 = 0.3481 \text{ cm}^2$$
$$D_5^2 = (0.336)^2 = 0.1129 \text{ cm}^2$$
$$D_{15}^2 - D_5^2 = 0.3481 - 0.1129 = 0.2352 \text{ cm}^2$$
$$\lambda = \frac{0.2352}{4 \times 10 \times 100} = \frac{0.2352}{4000} = 5.88 \times 10^{-5} \text{ cm}$$

Answer: The wavelength of light used is $\lambda = 5.88 \times 10^{-5}$ cm $= 588$ nm (or 5880 Å).

Question 5 [2 Marks]

In a Newton's rings experiment, the diameters of the 10th and 20th dark rings are found to be 3.20 mm and 4.50 mm, respectively, when air is present between the lens and glass plate. When a transparent liquid is introduced, the diameters of the same 10th and 20th dark rings become 2.74 mm and 3.85 mm, respectively. Determine the refractive index of the transparent liquid using Newton's rings method

Solution:

- **Given:**

$$\text{In Air: } D_{20} = 4.50 \text{ mm}, \quad D_{10} = 3.20 \text{ mm}$$
$$\text{In Liquid: } D'_{20} = 3.85 \text{ mm}, \quad D'_{10} = 2.74 \text{ mm}$$

- **Formula:**

$$\mu = \frac{(D_{20}^2 - D_{10}^2)_{\text{air}}}{({D'_{20}}^2 - {D'_{10}}^2)_{\text{liquid}}}$$

- **Calculation:**

$$(D_{20}^2 - D_{10}^2)_{\text{air}} = (4.50)^2 - (3.20)^2 = 20.25 - 10.24 = 10.01 \text{ mm}^2$$
$$({D'_{20}}^2 - {D'_{10}}^2)_{\text{liquid}} = (3.85)^2 - (2.74)^2 = 14.8225 - 7.5076 = 7.3149 \text{ mm}^2$$
$$\mu = \frac{10.01}{7.3149} \approx 1.3684$$

Answer: The refractive index of the liquid is $\mu \approx 1.37 \approx$ [R.I. of ethanol].